

Education

Bachelor of Science in Mechanical Engineering, Taylor University
Minors in Mathematics, Physics & Honors
GPA: 3.90/4.00, Dean's List

Anticipated Graduation: May 2027

Professional Experience

- **Mechanical Engineering Intern, Quantinuum** **May 2026-August 2026**
Worked as a mechanical design engineer focused on electronics packaging and photonics R&D at leading quantum computing company. Owned development of mechanical mounts for optical, electrical, and pneumatic components to enable increased photonics testing capabilities and test cadence. Designed mechanical components for electronics test setups, manufacturing processes, and commercial hardware deployment. Designed test rig for thermal strap qualification and characterization of thermal conductance as a function of bolt torque. Owned development of UHV test rig to verify components for future cryogenic and vacuum systems.
- **Undergraduate Researcher, Engineering Department, Taylor University** **February 2026-Present**
Developing thermal management solutions for an orbital robotics system. Responsible for development of robotic arm thermal model in MATLAB. Performing design, analysis, and TVAC testing of an actively actuated thermal louver. Researching and defining environmental, thermal, vacuum, and reliability requirements for robotic arm design sub-teams.
- **Mechanical Engineering Intern, Danner** **May 2025-August 2025**
Designed SolidWorks models and drawings for mounts of GNC/autonomy hardware and drivetrain subsystem components for off-road utility vehicles at electric vehicle startup. Developed an in-house mobile battery charger, participated in its fabrication, and troubleshoot assembly issues. Researched and selected parts and suppliers for future vehicle iterations. Created Jira workflow to manage engineering change requests to reduce lead time for engineering department.

Student Leadership & Activities

- **Hovercraft Competition** **February 2025-May 2025**
Responsible for design of thrust and lift subsystem components, using Fusion 360 and AutoCAD. Collaborated with electrical team members to develop hardware integration, wire harness routing, and component mounting solutions. Performed engineering analysis to optimize selection of motor/propellor combination and inform system design. Troubleshoot electrical and mechanical manufacturing and assembly issues. End result of competition was second place finish. At the conclusion of the project, gave a technical presentation.
- **Teaching Assistant, Physics & Engineering Department, Taylor University** **August 2024-Present**
Assisting professors with grading and facilitation/set up of labs. Leading review/teaching sessions as needed. Fostering a positive, productive, and engaging learning environment for students. Courses include University/General Physics I & II, Statics, Modern Physics (Study Table Leader), and Intro to Engineering.
- **Honors Guild Leadership Cabinet** **September 2023-Present**
Working alongside University Faculty, staff, and students to plan academic events for current Honors Guild students. Coordinating and facilitating academic speaking events for both university facility and guest speakers. Served as a student leader for study abroad in Italy (J-Term 2025).

Skills

- *Mechanical Design:* OnShape, Fusion 360, SolidWorks, AutoCAD, GD&T, PLM (Arena)
- *Simulation & Analysis:* Ansys Fluent, Autodesk FEA/CFD, EES, Hand Calcs
- *Programming:* MATLAB/Simulink, C/C++ Python, Mathematica, Excel, Jira
- *Manufacturing & DFM:* 3D Printing, CNC Machining, Welding, Laser/Plasma Cutting, Waterjet, Sheet Metal